HOSPITAL EQUIPMENT WIRING USING KRUSKAL'S ALGORITHM

1. Problem Statement

Develop a C++ program to represent hospital equipment units as vertices and possible wiring links as weighted edges. Apply Kruskal's algorithm to select the minimum-cost wiring connections. Display the selected edges and total wiring cost, and list the edges rejected because they create cycles.

2. Solution

The hospital equipment units are represented as vertices in a weighted graph, and each possible wiring connection is represented as an edge with a cost. Kruskal's algorithm sorts all edges in ascending order of cost and selects the cheapest edges that do not create cycles. A Disjoint Set Union (DSU) data structure checks whether the endpoints of an edge are already connected. If they are in different sets, the edge is selected; otherwise, it is rejected because it creates a cycle. The program then displays all selected wiring connections, the total minimum wiring cost, and the rejected edges.

3. Objectives

1. To represent hospital equipment units as vertices in a graph.
2. To represent possible wiring connections as weighted edges.
3. To implement Kruskal's algorithm in C++.
4. To find the minimum-cost wiring connections.
5. To calculate and display the total wiring cost.
6. To identify edges rejected because they create cycles.

4. Algorithm

7. Start the program.
8. Input the number of hospital equipment units and possible wiring connections.
9. Enter the name of each equipment unit.
10. Enter each wiring connection with its source, destination, and cost.
11. Sort all edges in ascending order of cost.
12. Initialize the Disjoint Set Union (DSU), with each vertex in a separate set.
13. Select the edge with the lowest cost.
14. If the edge connects two different sets, select it and merge the sets.
15. If both endpoints are already in the same set, reject the edge because it creates a cycle.
16. Repeat until all edges have been processed.
17. Display the selected edges and calculate the total minimum wiring cost.
18. Display the rejected edges that create cycles.
19. Stop the program.

5. Flowchart

The flowchart below illustrates the original Kruskal's algorithm process: input equipment and connections, sort edges, select edges without cycles, and display the selected edges, total cost, and rejected edges.

